JUNE, 2025

REGIONAL MARITIME UNIVERSITY

MARINE ELECTRICAL/ELECTRONIC ENGINEERING DEPARTMENT

BEE 27 / BME 27

END OF SEMESTER EXAMINATION

BEEE 207 – ROTATING ELECTRICAL MACHINES (ELECTRICAL MACHINES II)

INSTRUCTIONS: ANSWER QUESTION ONE (1) AND ANY FOUR (4) **TIME: 3 HOURS**

1(a) Describe the operation of Alternator or synchronous generator. **[5 Marks]**

(b) Describe the one important difference between d.c. generator and synchronous generator. **[5 Marks]**

(c) (i) Where in synchronous generator is the 3-phase armature winding placed? **[1 Mark]**

(ii) Outline two advantages of this arrangement in (i). **[2 Marks]**

(iii) What role does slip rings play in synchronous generator? **[2 Marks]**

(d) A 3-phase, 50 Hz induction motor has 8 poles. If the full-load slip is 2.5 %, determine (i) synchronous speed (ii) rotor speed and (iii) rotor frequency. **[5 Marks]**

2(a) (i) State the principle on which d.c. motors work. **[2 Marks]**

(ii) State two problems associated with the Armature control method used in the speed control of d.c. shunt motor. **[3 Marks]**

(b) A d.c. motor takes an armature current of 110 A at 480 V. The armature circuit resistance is 0.2 Ω. The machine has 6 poles, and the armature is lap-connected with 864 conductors. The flux per pole is 0.05 Wb. Calculate (i) the speed and (ii) the gross torque developed by the motor. **[5 Marks]**

3(a) Describe the principle of operation of 3-phase induction motor. **[8 Marks]**

(b) Explain slip as applied to three phase induction motor. **[2 Marks]**

4(a) A 3-phase induction motor having star-connected rotor has an induced e.m.f. of 50 V between the slip rings at standstill on open-circuit. The rotor has a resistance and reactance per phase of 0.5 Ω and 4.5 Ω respectively. Find the current per phase and the power factor at starting when (i) the slip rings are short-circuited (ii) the slip rings are connected to a star connected rheostat of 4 Ω per phase. **[8 Marks]**

(b) Comment on your answers in 4(a). **[2 Marks]**

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5 With aid of diagram, describe the starting method used in starting slip-ring induction motor and state two advantages slip-ring induction motors have over squirrel cage motors. **[10 Marks]**

6(a) With the aid of diagram, describe the speed and torque characteristic *(N/T_a)* of d.c. series motor. **[6 Marks]**

(b) Which type of d.c. motor will you chose for lathes, drills, and boring mills operations and why? **[4 Marks]**

7(a) With the aid of diagrams, explain the flux control method of speed control of d.c. shunt motor and state two advantages of this method. **[5 Marks]**

(b) A 4-pole, 500 V shunt motor has a total of 720 armature conductors which are lab connected. The full-load armature current is 60 A and the flux per pole is 3 mWb. The armature resistance is 0.2 Ω. The voltage drop across a brush is 1 V. Calculate the full-load speed of the motor. **[5 Marks]**

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